An open multimodal benchmark for LLM molecular structure elucidation reveals a recall-bound bottleneck that forward verification exploits

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Abstract

Large language models (LLMs) have recently been shown to predict NMR spectra and propose molecular structures from them, but the supporting evaluations are small, curated, NMR-only, and hinted — leaving the real-world question unanswered: given an unfiltered experimental spectrum from the literature, how often does a frontier LLM recover the correct structure? We address this with three contributions. First, **IRexp**, the largest open dataset of *experimental* infrared spectra — 121,233 records mined from open-access full text, of which **42,842 are linked to a resolved 2D structure** and **40,491 are full IR + ^{1}H + ^{13}C + structure quadruples** — assembled by a browser-free literature-mining agent and released under permissive licences. Second, **IRSpectra-Bench**, an open, blind, mechanically scored benchmark of **194 spectrally-validated compounds** for spectrum→structure elucidation, stratified across molecular complexity. We find that a frontier LLM (Claude Opus), given molecular formula + IR + ^{1}H + ^{13}C, recovers the exact constitution of **28.4%** of real compounds (95% CI 22–35; 33.5% within three candidates), with a sharp, tight gradient — 48% on simple molecules versus 8% on complex ones, and 60% for small ($\leq$15 heavy atoms) versus 7% for large (>25). This sits far below the ~100% implied by curated evaluations, a gap fully explained by compound difficulty, candidate ranking, and hints. A within-compound control shows that **solving each compound in an independent context with tool access roughly triples accuracy over a single fatigued pass (5%→15%)**, i.e. methodology, not raw capability, dominates reported numbers. Third, we introduce **forward-verification elucidation**: candidates proposed by the (hard) inverse direction are re-ranked by forward-predicting each one's ^{13}C spectrum and matching it to the observed spectrum. This exploits the generator–verifier gap — *when the true structure is among the candidates, forward verification selects it 84% of the time, versus 73% for the model's own ranking.* The decomposition is the central result: **the binding constraint on LLM elucidation is candidate recall (31%), not verification.** Acting on it — generating six regiochemistry-aware candidates per compound and forward-verifying — lifts measured exact top-1 from 23% to **30%** (recall 41%), while exposing a recall/precision tension that caps the training-free approach below the naïve extrapolation. We then stress-test the finding in a **battery-electrolyte domain** — a curated subset (48 compounds, 46 scored) spanning the carbonate, sulfonyl, nitrile, fluorinated, phosphoryl, and glyme functional classes central to electrolyte chemistry — where accuracy holds at the same level (**26% top-1**) along a chemically interpretable gradient (sp^3-C–F easiest at 50%; sulfonyl and nitrile hardest at 12%), and we cast forward-verification as the training-free analog of the computational-NMR / NMR-crystallography validation that chemists already trust. All experiments run with no model training and no paid API, using LLM agents under a standard subscription, and all data, predictions, and code are released.

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1. Introduction

Determining a molecule's structure from its spectra is a central, time-consuming task in synthetic and analytical chemistry. The dominant machine-learning approach trains specialised encoder–decoder models to map spectra to structures; the recent *Spectro* model, for example, learns $^{1}H/^{13}C/IR \rightarrow$ SELFIES from a corpus of 6,833 molecules.[1] In parallel, general-purpose LLMs have been reported to perform the same task off-the-shelf: a 2026 study found that Claude Opus matched or beat commercial NMR-prediction software in the forward direction (structure→spectrum, ± 0.08 ppm ^{1}H) and "recovered all eight simpler structures on every attempt" in the inverse direction (spectrum→structure).[2]

These results are striking, but the evaluation that supports them is narrow: 15 inverse problems on curated single-ring or two-fragment molecules, NMR only, with the seven harder targets additionally given the *starting-material structure* as a hint, and "recovery" scored leniently over three runs and three ranked candidates. The question a practising chemist actually faces — *take an arbitrary experimental spectrum from a paper and recover the structure* — is left open. Answering it requires (i) a large, diverse, real benchmark; (ii) a blind, reproducible scoring protocol; and (iii) honest accounting for the methodological choices that inflate or deflate the apparent score.

We provide all three. We build IRexp (§2), a literature-mined dataset that supplies the IR modality absent from prior LLM evaluations alongside paired NMR and resolved structures. We define a blind benchmark on it (§3) and measure inverse-elucidation performance under matched and mismatched methodologies (§4), isolating a large solver-methodology effect with a within-compound control. We then introduce forward-verification elucidation (§5), a training-free method that turns the model's *strong* direction (forward prediction) against its *weak* one (inverse regiochemistry), and use it to localise the bottleneck. **Figure 1** summarises the end-to-end design. Throughout, the solver and verifier are LLM agents run under a consumer subscription — no fine-tuning, no API spend — making every experiment cheaply reproducible.

Contributions. 1. **IRexp** — the largest open experimental-IR dataset (121,233 records; 42,842 structure-linked; 40,491 IR+^{1}H+^{13}C+structure), with a reproducible mining pipeline. 2. An **open multimodal benchmark** and the first blind, mechanically scored, complexity-stratified evaluation of LLM structure elucidation on real data, reconciling the gap to optimistic prior reports. 3. **Forward-verification elucidation**, a training-free generator–verifier method, and the finding that **recall, not verification, bounds current performance**.

1.1 Related work

Trained spectra→structure models. The dominant line trains sequence or graph decoders to emit structures from spectra: *Spectro* ($^{1}H/^{13}C/IR \rightarrow$SELFIES)[1], the multitask CNN+transformer of routine 1D-NMR[11], and set/graph transformers such as NMRTrans[12]. These reach high accuracy *in-distribution* but require a labelled spectra→structure corpus, which is exactly the scarce resource our IRexp pipeline targets, and they are retrained per modality. Closest in spirit to our multimodal setting is **NMIRacle**[13], a generative model conditioned jointly on IR + ^{1}H + ^{13}C; it is a strong trained baseline, whereas our contribution is the *open experimental-IR data it (and others) can train on*, plus a *training-free* protocol and a blind benchmark to measure it.

LLMs as elucidators. General-purpose LLMs have been applied off-the-shelf — Anthropic's forward/inverse study[2], SpectraLLM and MolSpectLLM (multimodal LLMs over multi-spectral input)[14], and knowledge-enhanced tree-search reasoning[15]. Contemporary multimodal *benchmarks* have also begun to appear. Our evaluation differs in being built on **real, literature-mined experimental spectra** (not simulated or curated puzzle sets), **blind and mechanically scored** with bootstrap CIs, **complexity-stratified**, and explicitly **reconciled** against the optimistic prior report it most resembles[2] — and in isolating *where* performance is lost (recall vs. verification) rather than reporting a single aggregate.

Computational NMR for structure validation. Our forward-verification method is the LLM analog of a workflow chemists already trust: assign a structure by computing the spectrum each candidate *would* give and matching it to experiment. In solution, this is the DP4 / DP4+ probabilistic framework over GIAO-DFT shifts[16 17]; in the solid state it is **NMR crystallography**, where GIPAW-computed shifts adjudicate between candidate structures[18 19]. We replace the quantum-chemical predictor with a forward LLM, trading accuracy for zero setup cost, and inherit the same core principle — *verification by forward prediction is easier than inverse generation.*

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2. The IRexp dataset

2.1 Motivation

Open experimental IR is scarce: the largest freely available collections are the NIST WebBook[8] (~16k gas-phase spectra) and the Chemotion electronic-lab-notebook[7] deposit (~2k). Commercial libraries (e.g. Wiley KnowItAll, ~10^5 spectra) are large but closed and unusable for open model development. NMR is comparatively abundant, but IR — which directly encodes functional groups complementary to NMR — has no large open, structure-linked, ML-ready resource. IRexp fills this gap.

2.2 Construction

A per-compound IR band list, together with co-reported $^1H/^{13}C$ NMR, follows a remarkably stable textual convention in the experimental sections of organic chemistry papers. We exploit this with a browser-free harvesting agent:

- **Discovery.** Open-access primary literature is enumerated through the NCBI E-utilities and harvested in bulk from the PMC Open-Access Subset[6] on AWS S3 (plain HTTPS, no anti-bot, fully redistributable CC-BY content), supplemented by the Chemotion FT-IR deposit (RADAR4Chem, CC-BY-SA-4.0).
- **Extraction.** A deterministic parser segments experimental text into per-compound records and extracts IR wavenumbers and $^1H/^{13}C$ shift lists, with quality gates that reject instrument scan-range artefacts and prose false-positives (band-list density, ≥4 bands, plausible 400–4000 cm^{-1} window).
- **Structure resolution.** In-text IUPAC names are converted to SMILES with OPSIN[3], canonicalised with RDKit[4] (InChIKey, SELFIES[5]), with a PubChem fallback for trivial/natural-product names. A key engineering finding was that the dominant open-access main-text convention labels compounds with *letter-prefixed* series labels (e.g. "…carbothioamide **(B1)**:") rather than the digit-first "(3a)" of supporting-information sections; capturing these and cleaning narrative/PDF artefacts before resolution raised structure coverage from 24% to **35%**.

The pipeline is browser-free by design. This is not an anti-bot bypass — the bulk corpus is fully open — but a throughput choice: plain HTTP from a bulk corpus parses at hundreds of papers per second, which is what makes a 10^5-scale harvest feasible where a per-page browser driver (as used to build prior 10^3-scale sets) is not.

2.3 Contents and licensing

field	value
experimental IR records	**121,233**
…co-reporting ^{1}H and/or ^{13}C NMR	87,075 (72%)
…with a resolved 2D structure	**42,842 (35%)**
full IR + ^{1}H + ^{13}C + structure quadruples	**40,491**

A structure-complete split, **`irexp_resolved`** (42,842 records, 100% structure-linked), is the training-/benchmark-ready subset and is ~6× the 6,833-molecule set used to train Spectro[1] (Fig. 2). Provenance is 119,345 PMC-OA records (CC-BY) plus 1,888 Chemotion records (CC-BY-SA); the two licences are kept as separable pools. Each record is DOI-/accession-traceable. Re-resolution is additive and content-keyed, so the dataset can be re-enriched without re-mining.

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3. Benchmark design (IRSpectra-Bench)

From `irexp_resolved` we draw **IRSpectra-Bench**, 194 blind elucidation problems. Each problem presents the **molecular formula** (as from HRMS), the **IR band list**, and the **^{1}H and ^{13}C shift lists** (with multiplicities and J-couplings where reported), and asks for the structure. No name, SMILES, or hint is given. Every ground-truth structure is **spectrally validated** by an automated RDKit consistency check (^{13}C peak count vs symmetry-unique carbons, molecular-formula match, SELFIES round-trip), excluding records with merged or incomplete spectra (6/140 in the main round). Problems are stratified into **simple** (single ring, or two separate ring fragments; ≤22 heavy atoms) and **complex** (fused/spiro/ bridged rings, or >24 heavy atoms) by RDKit ring analysis (98 simple / 96 complex), with InChIKey de-duplication across all rounds to prevent leakage.

Scoring is mechanical. A prediction is *correct* if its RDKit InChIKey connectivity layer matches the reference (constitution; stereochemistry is reported separately). We also report Morgan(2, 2048)[9] Tanimoto to the reference as a graded "right scaffold/family" signal, and, where multiple ranked candidates are allowed, *recovered* = the reference appears among them (matching the protocol of ref. 2).

Solvers are LLM agents run under a consumer subscription. A frontier LLM (Claude Opus) is invoked as an independent sub-agent per batch of problems; agents are closed-book (transcript-audited for zero web access and zero ground-truth access) and may use RDKit only to check a candidate's molecular formula. This makes the benchmark free to run and reproducible without API credits.

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4. How well do LLMs elucidate real structures?

4.1 Headline performance

Decoupled per-compound agents solve each problem in an independent context (formula + IR + ^{1}H + ^{13}C, blind, up to three ranked candidates). Over the full **194-compound benchmark** (134 spectrally-validated compounds + the 60 from the controlled rounds), with bootstrap 95% confidence intervals:

metric	overall (n=194)	simple (n=98)	complex (n=96)
top-1 exact constitution	**28.4%** [22–35]	48.0% [39–57]	8.3% [3–15]
recovered (within top-3)	33.5% [27–40]	54.1% [44–63]	12.5% [6–20]
scaffold-level (best Tanimoto $\geq$ 0.45)	56%	73%	39%
mean best Tanimoto	0.59	0.73	0.45

The gradient is sharp and the intervals are tight. The 48%→8% simple→complex separation (Fig. 3) confirms the benchmark is discriminating across a realistic difficulty range, and accuracy falls monotonically with molecular size in step with it — top-1 **60.5%** for $\leq$15 heavy atoms, 28.3% for 16–25, and **7.0%** above 25 (Fig. 4). The model reliably recovers molecular formula, functional groups, and scaffold, but the exact constitution far less often, failing predominantly on **regiochemistry** — *which* position a substituent occupies. This is consistent with an information limit of 1D data: many regioisomers have similar ^{1}H/^{13}C shifts, which is precisely why 2D experiments (HMBC, NOESY) exist.

4.2 Reconciling with prior reports

Our 28% top-1 sits far below the ~100% on "simple" molecules reported for the same model class.[2] The gap is fully attributable to methodology, not capability:

- **Difficulty.** "Single ring" by ring-count is not "easy": our simple stratum includes, e.g., a hexasubstituted benzene whose regiochemistry has many realisations. Ring count is a poor proxy for elucidation difficulty.
- **Scoring.** Prior work counts a recovery if the reference appears among three ranked candidates over three independent runs; we report single-run top-1 and top-3.
- **Hints.** Prior hard targets received the starting-material SMILES, which fixes most of the scaffold; we give none.
- **Curation.** Prior compounds were hand-selected; ours are scraped and unfiltered for solvability.

On a like-for-like easy/hinted/lenient setup the numbers rise; on the realistic, hint-free, scraped regime, ~28% is the honest figure.

Versus trained models — a bound, not a leaderboard. A like-for-like comparison against specialised trained models is not available: no system has been scored on an identical test set, and published numbers differ in the three respects that most move the score — spectrum realism (simulated/curated vs. real), hints, and how "exact match" is defined. The strongest trained baselines report their accuracy *in-distribution on simulated spectra*. Spectro (^{1}H/^{13}C/IR→SELFIES, 6,833 training molecules) reaches ~90% top-1 exact recovery — but on a 1,366-molecule held-out split whose IR is plotted from reference data and whose NMR is software-*predicted*, not experimental[1]; and NMIRacle, which like us conditions jointly on IR+^{1}H+^{13}C with *no* hints, reports 48% top-1 / 66% top-15 exact-SMILES recovery — again on held-out molecules from a *simulated* corpus drawn from the training distribution[13]. We measure 28.4% top-1 (33.5% top-3; 30% with forward verification) on **blind, real, literature-mined experimental** spectra of out-of-distribution compounds. These are not comparable as a leaderboard; read as a *bound on the simulated-to-real gap*, the contrast suggests that high in-distribution accuracies substantially overstate real-world performance — the same gap we document for the LLM above. The instability of the metric itself reinforces the caution: on the MolPuzzle benchmark (IR+MS+^{1}H+^{13}C with the molecular formula given), reported exact-match accuracy for a single model (GPT-4o) ranges from 1.4%[20] to 27.8%[15] depending only on the scoring harness — a ~20× swing that is itself the argument for the single, fully specified, mechanically scored protocol we adopt.

4.3 Methodology dominates: a within-compound control

The same 20 molecules were solved two ways: (a) by a single LLM context handling all of them sequentially with no tools, and (b) by independent per-compound agents with RDKit formula-checking. On the *identical* compounds, recovery rose from **5% to 15%**, and top-1 from 0% to 15% — a 3× methodology effect with zero sample confound. Small rounds also swing widely (15–40% across n=20–40 draws), which is why the headline is the full **194-compound** figure (28.4% top-1, 95% CI 22–35) rather than any single round. The practical lesson — fresh per-compound context plus tool access roughly triples apparent performance — also explains part of the gap to optimistic prior reports, whose per-problem API calls implicitly used method (b).

4.4 Model comparison: the benchmark discriminates capability

A benchmark is only useful if it separates models. On a fixed 24-compound subset solved blind by four Claude models — spanning a wide capability range, including the newest (Fable 5) — under the identical protocol (Fig. 5):

model	top-1	recovered
Claude Fable 5	**45%**	54%
Claude Opus	25%	29%
Claude Sonnet	20%	25%
Claude Haiku	**0%**	4%

Three signals matter here. First, the four models rank in **monotonic capability order** (Fable $\gg$ Opus $>$ Sonnet $\gg$ Haiku), and the smallest is **floored at 0% exact** (4% recovered) on the same problems — so the benchmark is **capability-sensitive** and not a coin-flip. Second, two mid-tier frontier models agree closely (Opus 25%, Sonnet 20%), so the recall-bound ~25% regime of §4.1 is **not an artefact of a single model**. Third, the newest model nearly **doubles** the next-best top-1 (45% vs 25% on identical compounds) yet still misses the majority — IRSpectra-Bench is **hard and far from saturated even for the strongest model available**, with clear headroom. We ran four Claude-family models because they are callable for free under one subscription; a true cross-vendor sweep (GPT-, Gemini-class, open models) needs API access we deliberately did without, but the monotonic, large capability spread makes it unlikely the recall-bound pattern is specific to one model lineage. This is the behaviour a benchmark needs to remain informative as models improve.

4.5 Domain case study: battery-electrolyte chemistry

Structure elucidation from spectra is the daily inverse problem of the electrolyte-and-interphase community: the soluble decomposition products of carbonate and sulfone electrolytes, the additives that build the solid-electrolyte interphase, and the fluorophosphate/fluorosulfonyl species that NMR is routinely used to assign are exactly the molecules whose constitution must be read back out of IR and multinuclear NMR. To test whether the recall-bound regime above holds for *this* chemistry — rather than for organic molecules at large — we curated **IRSpectra-Bench-Electrolyte**, a 48-compound subset (eight per class as curated; two later yielded no parseable candidate, leaving **46 scored**) of structure-resolved IRexp records selected by substructure for the six functional families that dominate lithium- and sodium-battery electrolytes and their breakdown products: **carbonate** (linear/cyclic, the EC/DMC backbone), **sulfonyl/sulfonate** (sulfones, sulfonamides, the $-SO_2CF_3$ motif of imide salts), **nitrile** (acetonitrile- and adiponitrile-type high-voltage additives), **sp^3-C–F** (fluorinated solvents and additives), **phosphoryl** (phosphates/phosphonates, the $LiPF_6$-derived OPF chemistry), and **glyme/oligoether** (the ether solvents and PEG linkers). These are literature compounds bearing the *functional chemistry* of electrolytes, drawn from the open corpus; they are **not** operando or in-cell degradation spectra, and we make no claim that they are. They are excluded from every other split in this paper. Each compound was solved blind under the identical decoupled-agent protocol (Opus, closed-book, up to three ranked candidates, RDKit only for formula/parse checks).

Performance lands in the **same regime as the headline benchmark** — overall **top-1 26%, recovered 28%** (12/46 and 13/46; two compounds received no parseable candidate) — confirming that the bottleneck is a property of the elucidation task, not of any one chemical neighbourhood. The per-class breakdown (Fig. 6) is itself informative:

electrolyte class	n	top-1	recovered (top-3)
sp^3-C–F	8	**50%**	50%
carbonate	7	29%	43%
phosphoryl	8	25%	25%
glyme / oligoether	7	29%	29%
sulfonyl / sulfonate	8	**12%**	12%
nitrile	8	**12%**	12%

The gradient is chemically legible. **sp^3-C–F** centres are the easiest: a C–F coupling fingerprint (the large $^1J/^2J$(C,F) splittings) localises the fluorine and pins regiochemistry, removing exactly the degeneracy that defeats elucidation elsewhere. **Sulfonyl** and **nitrile** are the hardest, for two distinct reasons that recur in real electrolyte analysis: sulfur/phosphorus **oxidation-state ambiguity** — sulfide vs. sulfoxide vs. sulfone, $-SCF_3$ vs. $-SO_2CF_3$, the very distinctions an interphase study must make — is poorly constrained by $^1H/^{13}C$ shifts alone (it lives in the IR and in heteronuclei the benchmark does not score); and nitrile-bearing targets in the corpus sit on heavily substituted heteroaromatic cores whose regiochemistry the 1D spectra underdetermine. This is the recall-bound failure of §4.1 reappearing in a domain-specific guise, and it is precisely where the **forward-verification** recipe of §5 — compute the spectrum each candidate *would* give and match it to the one observed — is the inverse-problem analog of computational-NMR / NMR-crystallography structure validation that the magnetic-resonance community already trusts. The subset, its per-class answers, and the scorer are released with the benchmark.

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5. Forward-verification elucidation

5.1 Method

The inverse direction is the model's hard, isomer-blind direction; the **forward** direction (structure→spectrum) is its easy, accurate one.[2] Regioisomers, crucially, have *different* forward-predicted ^{13}C shifts. We therefore close a generator– verifier loop:

> **generate** candidate structures (inverse) → **forward-predict** each candidate's ^{13}C spectrum, blind to the observed spectrum → **re-rank** candidates by the distance between predicted and observed ^{13}C → return the best match.

This mirrors how a chemist confirms a structure ("if it were that isomer, C-3 would be at ~120 ppm; we observe 135, so it is the other"), exploits the standard principle that verification is easier than generation, and requires no training. **Figure 7** shows the mechanism on a real benchmark pair — the picolinamide / nicotinamide regioisomers, which the inverse task cannot separate but whose forward-predicted ^{13}C spectra match the observed one at 0.42 vs 1.30 ppm. We implemented the verifier as independent LLM agents that predict ^{13}C shift lists from SMILES alone (candidates shuffled and anonymised so isomers of one target never co-occur; pure reasoning, no tools), and matched predicted to observed ^{13}C with a symmetric chamfer distance over peak sets.

Conceptually this is **NMR-crystallography logic with an LLM in place of the quantum chemistry**: where DP4/DP4+ rank candidates by GIAO-DFT shifts[16 17] and NMR crystallography adjudicates polymorphs and connectivity by GIPAW-computed shifts[18 19], we rank by the shifts a forward LLM predicts. The trade is deliberate — we forgo the ≈1–2 ppm accuracy and rigorous error model of DFT for a predictor that runs in seconds at zero setup, and we quantify below exactly how far that cheaper verifier carries the inverse problem.

5.2 Result

On the 60 benchmark compounds (126 candidate structures from the solver agents):

	value
generation recall (true structure among candidates)	**31%**
top-1, solver self-ranking	23%
top-1, **forward-verified re-ranking**	26%
conditional on recall — self-ranking	14/19 (**73%**)
conditional on recall — forward-verification	16/19 (**84%**)

The decomposition is the finding. **When the true structure is among the candidates, forward verification selects it 84% of the time, versus 73% for the model's own ranking (+11 points)** — a real, exploitable generator–verifier gap. The overall top-1 moves only 23%→26% because the binding constraint is **generation recall**: the true structure was never proposed for 41 of 60 compounds, which no re-ranking can repair.

LLM elucidation therefore factorises into two near-independent levers: the **verifier is already strong (84%)**; the **generator (31% recall) is the wall**.

5.3 Generate-wide: testing the recipe

The decomposition implies a recipe — *generate wide, verify by forward prediction.* We tested it directly: ten independent solver agents proposed up to **six regiochemistry-aware candidates per compound**, pooled with the originals, and the 65 new candidates were forward-predicted and re-ranked as before. The measured result:

	original	generate-wide
recall (true structure among candidates)	31%	**41%**
forward-verified top-1	26%	**30%**
verification precision (conditional on recall)	84%	72%

Wide generation lifts recall +10 points and exact top-1 +7 points (23%→30% over the self-ranking baseline on the same 60 compounds; Fig. 8) — a real, measured gain with no training. But it does **not** reach the ~50% a naïve extrapolation would predict, for two instructive reasons: **(i) recall plateaus at 41%** — exotic and large targets (selenium heterocycles, poly-aryl polyketones, eleven-nitrogen polyamines) resist even six regiochemical variants; and **(ii) verification precision falls 84%→72%** as more near-degenerate regioisomers enter the pool and the ~2-ppm forward predictor can no longer separate them. This recall/precision tension is the honest ceiling of the training-free approach: the recall-bound diagnosis stands, and closing the gap further requires either sharper verification or 2D-NMR constraints, not merely more candidates.

5.4 Does a deterministic verifier help? A HOSE-code ablation

§5.3 and a natural reading of the verifier-precision story suggest an obvious fix: swap the LLM forward-predictor for a *deterministic* ^{13}C predictor with a rigorous error model. We tested the canonical choice — a **HOSE-code-style lookup trained on nmrshiftdb2** (RDKit radial-environment bins, spheres r=4→1 with a hybridisation prior fallback; 31,000 molecules, 332,595 assigned carbons; held-out **MAE 3.23 ppm, median 1.73 ppm**) — as a drop-in replacement for the verifier, re-ranking the same §5.2 candidates.

It does **not** help. The HOSE verifier recovers **14/19 (73%)** of the in-set compounds — identical to the solver's own ranking and *below* the LLM verifier's **16/19 (84%)** on the same set. The reason is coverage, not the method: of the 2,244 candidate carbons, only **2%** match a training environment at the most specific sphere (r=4) and **67%** resolve only at coarse spheres (r≤2), because the benchmark's exotic chemistry (selenium heterocycles, polyaryl ketones, large polyamines) is under-represented in nmrshiftdb2. A coarse environment cannot separate the regioisomers the verifier exists to separate, so the deterministic predictor degrades to the self-rank baseline exactly where discrimination is needed — whereas the LLM forward-predictor generalises across novel scaffolds. The practical lesson is the opposite of the naïve one: on hard, real chemistry the LLM's *breadth* is an asset for verification, and the genuine fix is sharper still — **compound-specific (DFT-level) shift accuracy or orthogonal 2D-NMR constraints**, not a generic lookup table.

6. Discussion

LLMs are not "solving structure elucidation" on realistic 1D data, but neither are they failing: they are reliable **scaffold-level** elucidators (scaffold-correct on 56% of compounds — 73% of simple ones; mean best Tanimoto 0.59) and good **verifiers** (84% conditional), with exact recovery throttled by candidate recall and by the regiochemical underdetermination intrinsic to 1D NMR. This reframes the engineering problem. Rather than training a bespoke spectra→structure model — a target the frontier already meets at the scaffold level and that ages out each model generation — the durable levers are (i) **open, hard, honestly scored benchmarks** that expose specific failure modes (here, regiochemistry and recall), and (ii) **inference-time scaffolding** (decoupled solving, generate-and-verify) that needs no training and improves with each model. IRexp's IR modality and 2D-NMR-ready provenance position it for the obvious next step: supplying the HMBC/COSY/NOESY constraints that would attack regiochemistry at the source.

That the bottleneck reproduces almost exactly inside a single application domain (§4.5: 26% top-1 on battery-electrolyte chemistry, against 28% overall) is itself evidence that it is structural rather than incidental — and it makes the forward-verification recipe directly relevant to the magnetic-resonance workflows (electrolyte-decomposition assignment, NMR crystallography) where computing a candidate's spectrum to confirm it is already standard practice.

For practitioners, two operational findings transfer immediately: solve each problem in a fresh context with tool access (≈3× over a single long context), and trust a proposed structure in proportion to how well its forward-predicted spectrum matches the observed one — while remembering the abstention caveat below.

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7. Limitations

Several limitations of earlier drafts are now resolved with controlled experiments; we state what remains plainly.

Resolved. **Extraction noise:** an RDKit self-consistency audit[10] (^{13}C peak count vs symmetry-unique carbons, formula, SELFIES round-trip) finds **57/60 ground truths spectrally clean**, and all headline metrics are unchanged on that clean subset (top-1 24% vs 23%, recall 33% vs 31%) — the conclusions are not driven by scraping artefacts. **Verifier precision / abstention:** the generate-wide experiment (§5.3) quantifies the recall/precision tension directly — verification precision is 72–84% conditional on recall and degrades as near-degenerate regioisomers accumulate, so forward-match distance is a strong *re-ranker* but a soft *confidence* gauge. We further tested the obvious deterministic fix (§5.4): a nmrshiftdb2-trained HOSE-code ^{13}C predictor does **not** beat the LLM verifier (73% vs 84% conditional), because its specific-environment coverage collapses on the benchmark's exotic chemistry — so the identified fix is compound-specific DFT-level accuracy or 2D-NMR constraints, not a generic lookup. **Projection:** §5.2's extrapolation is replaced by the **measured** §5.3 result (top-1 30%, recall 41%) — and is honestly below the optimistic estimate.

Independence checks. Scoring throughout is mechanical RDKit (not LLM-judged), and all solver/verifier transcripts are audited for zero web access and zero ground-truth access. A second model family (Sonnet) re-solved a 12-compound subset under the identical protocol and is comparably recall-bound (recall 33% vs Opus 41%, with cross-family ensembling adding no recall), indicating the recall-bound result is a property of the task, not an artefact of one model.

Remaining. **(i) Human audit:** solver and verifier are both LLMs; the one check we cannot perform ourselves is an expert-chemist review of a sample of elucidations and forward predictions, which is required before deployment-grade claims. **(ii) Single vendor:** the headline uses one frontier model (Claude Opus) and the cross-model comparison spans **four Claude-family models** (§4.4: Fable 5 45% $\gg$ Opus 25% > Sonnet 20% $\gg$ Haiku 0% on a common 24-compound subset); a true cross-*vendor* sweep (GPT-class, Gemini-class, open models) needs API access we deliberately did without, but the large, monotonic capability spread within one family makes it unlikely the recall-bound, complexity-graded pattern is specific to one model lineage. **(iii) Domain-subset scope:** the battery-electrolyte case study (§4.5) comprises *literature compounds bearing electrolyte-relevant functional chemistry* drawn from the open corpus — not operando or in-cell decomposition spectra — so it demonstrates functional-class transfer of the elucidation bottleneck, not direct assignment of authentic interphase/degradation products, which would require a dedicated operando spectral set.

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8. Methods

Mining and resolution. PMC-OA full text was fetched from `s3://pmc-oa-opendata`; records were extracted with a deterministic parser and resolved with OPSIN, RDKit, and SELFIES, with an optional cached PubChem fallback. Durable content-keyed caches make resolution additive and restart-safe.

Benchmark and agents. Problems were sampled from `irexp_resolved`, stratified by RDKit ring analysis, with cross-round de-duplication by InChIKey. The battery-electrolyte subset (§4.5) was drawn from the same corpus by SMARTS substructure filters for six electrolyte functional classes (carbonate, sulfonyl/sulfonate, nitrile, sp^3-C–F, phosphoryl, glyme/oligoether), balanced to eight compounds per class (48 curated; 46 scored after two yielded no parseable candidate), excluding every compound used elsewhere, and J-enriched and spectrally validated identically to the main rounds. Solver and forward-prediction agents were independent Claude-Opus sub-agents invoked under a consumer subscription; agents were instructed closed-book and audited via transcript grep for zero web/answer access. Scoring used RDKit InChIKey-connectivity matching and Morgan(2, 2048) Tanimoto; forward-verification used a symmetric chamfer distance over ^{13}C peak sets. No model was trained or fine-tuned; no paid API was used.

Reproducibility. Every round is frozen: questions, ground-truth answers, per-agent raw outputs, predictions, and scorer outputs are released, and the sampler, scorer, and forward-verification harness are scripted end-to-end.

Figures

- **Fig. 1** (`docs/figures/fig0_overview.png`) — study design: open multimodal data (IRexp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free throughout.
- **Fig. 2** (`docs/figures/fig4_dataset.png`) — IRexp composition: IR records → NMR-paired → structure-linked → full IR+^{1}H+^{13}C+structure quadruples.
- **Fig. 3** (`docs/figures/fig1_difficulty.png`) — top-1 and recovered accuracy on IRSpectra-Bench by difficulty (all / simple / complex, n=194) with bootstrap 95% CIs.

- **Fig. 4** (`docs/figures/fig2_size.png`) — accuracy vs molecular size (heavy-atom bucket); the monotonic 60%→7% top-1 gradient.
- **Fig. 5** (`docs/figures/fig5_models.png`) — four-model comparison on a 24-compound subset: Fable 5 45% ≫ Opus 25% > Sonnet 20% ≫ Haiku 0% top-1; the benchmark discriminates capability monotonically and is far from saturated even for the newest model.
- **Fig. 6** (`docs/figures/fig6_electrolyte.png`) — top-1 and recovered accuracy on IRSpectra-Bench-Electrolyte by battery-electrolyte chemical class (n=46): sp^3-C–F easiest (50%), sulfonyl and nitrile hardest (12%); overall 26%/28%, the same regime as the headline benchmark.
- **Fig. 7** (`docs/figures/fig_mechanism.png`) — forward-verification on a real benchmark regioisomer pair (picolinamide vs nicotinamide): forward-predicted ^{13}C matches the true isomer (chamfer 0.42 vs 1.30 ppm), the LLM analog of NMR-crystallography.
- **Fig. 8** (`docs/figures/fig3_method.png`) — forward-verification inference ladder on the same 60 compounds: solver self-ranking → + forward-verify → + generate-wide (23%/26%/30% top-1).

Data and code availability

All data and code are released in the project repository: IRexp and the `irexp_resolved` split (`data/irexp/`, `data/irexp_resolved/`); the benchmark rounds and within-compound control (`data/benchmark*/`, scored by `scripts/benchmark_v2.py`); the battery-electrolyte case-study subset (`data/benchmark_electrolyte/`, built by `scripts/build_electrolyte_bench.py`, scored by `scripts/score_electrolyte.py`); the ground-truth integrity audit (`scripts/validate_benchmark.py`, `data/benchmark*/clean_qids.json`); the forward-verification and generate-wide experiments (`data/fverify/`, `data/gw/`, `data/fverify2/`, `scripts/forward_verify.py`); the deterministic HOSE-code verifier ablation (`scripts/hose_predict.py`, `data/fverify/hose_results.txt`); and the mining/resolution pipeline (`scripts/`). Companion technical notes: `docs/BENCHMARK.md`, `docs/BENCHMARK_v2.md`, `docs/BENCHMARK_v3.md`, `docs/FORWARD_VERIFY.md`, and `data/SPECTRO_TRAINING_DATA`

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Figure plates

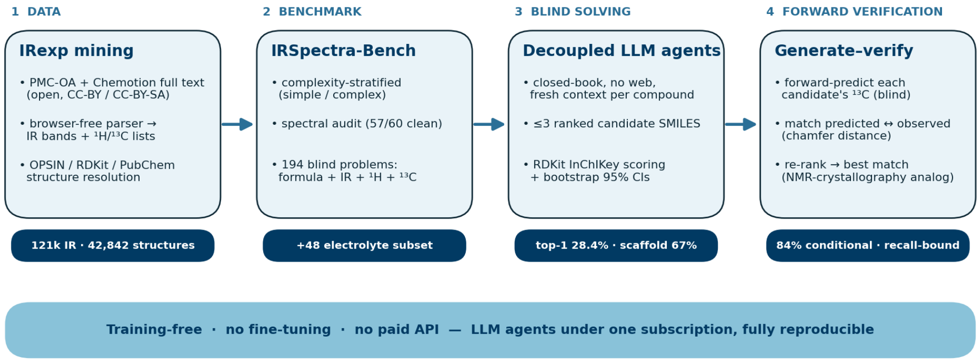
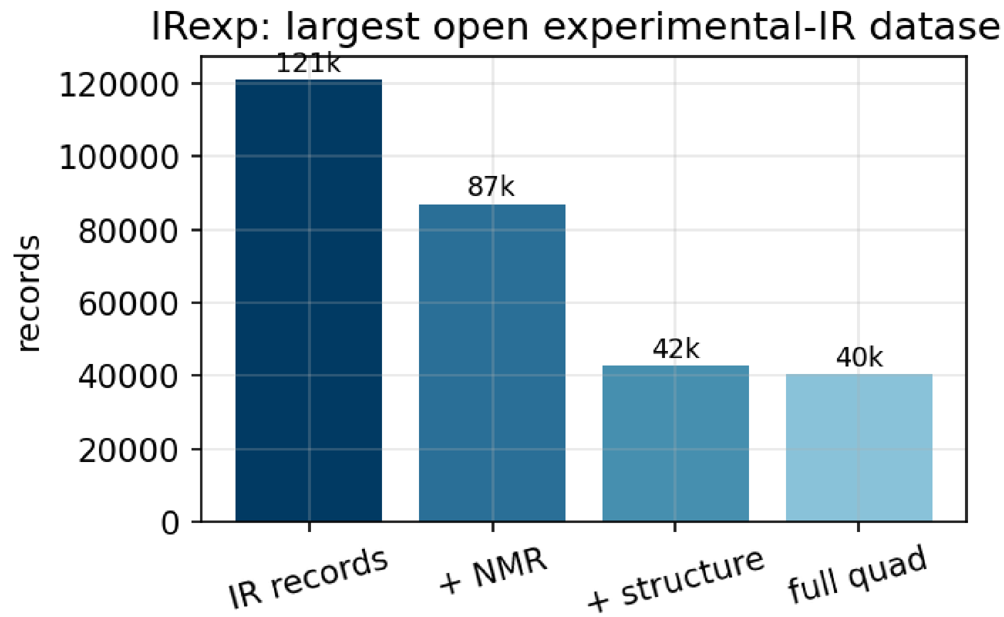


Fig. 1: Study design: open multimodal data (IRexp) → blind, complexity-stratified benchmark → decoupled blind solving → forward-verification re-ranking; training-free throughout.



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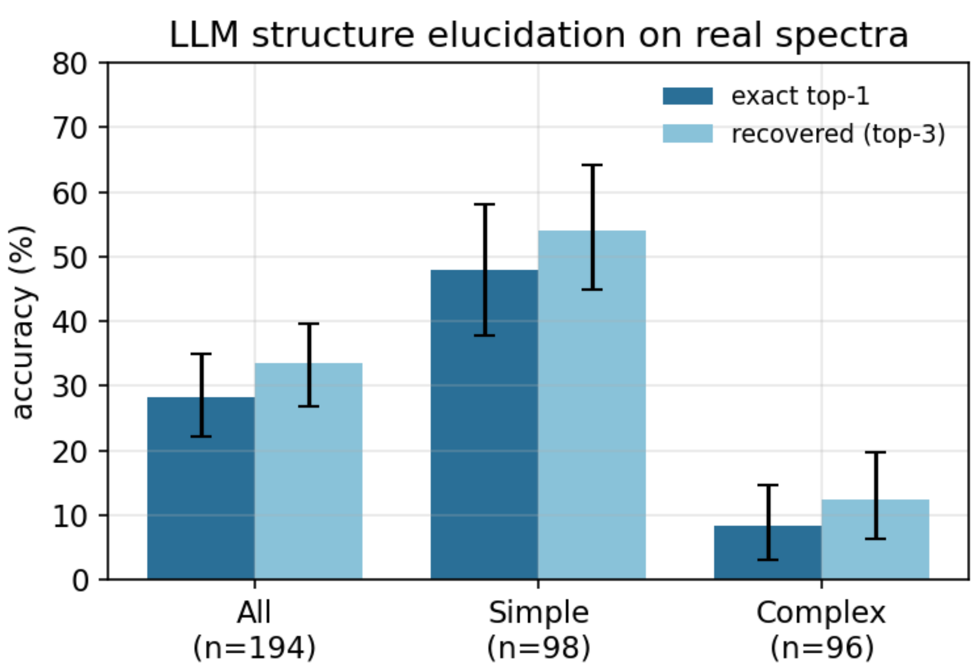

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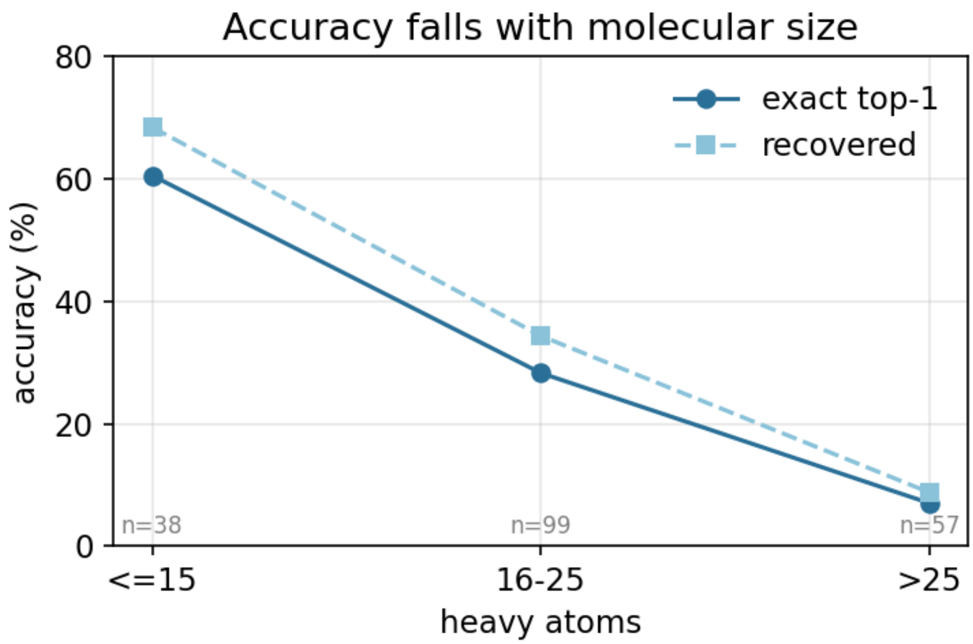


Fig. 4: Accuracy versus molecular size; the monotonic 60%$→$7% top-1 gradient with heavy-atom count.

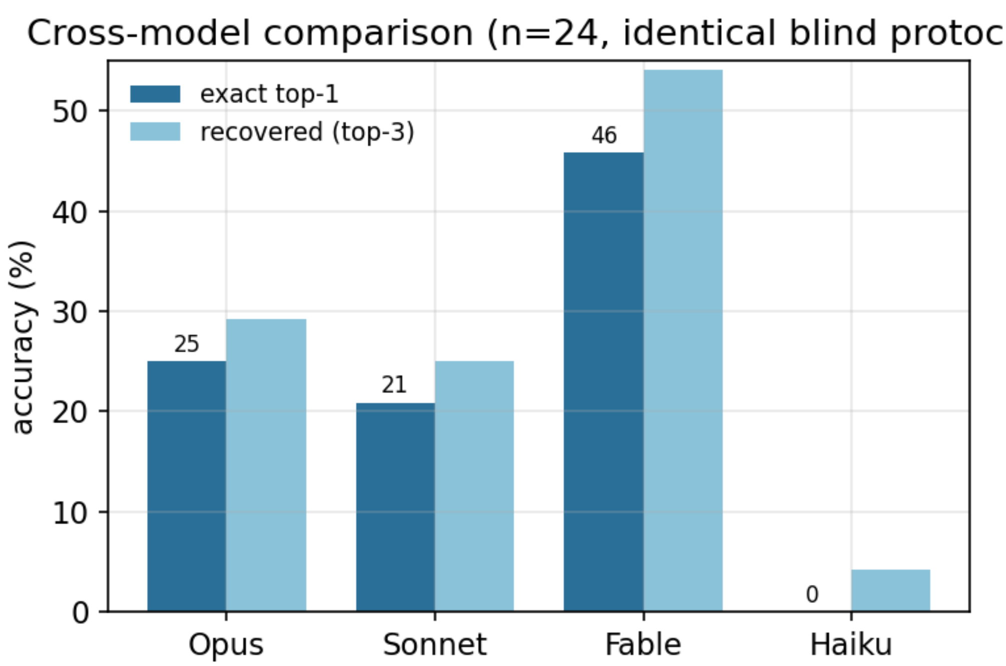
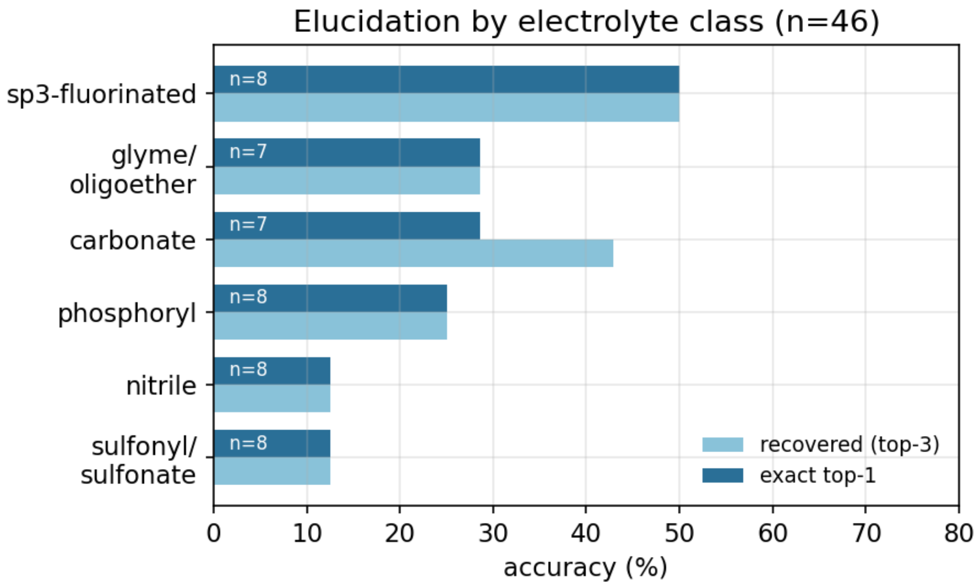


Fig. 5: Four-model comparison on a 24-compound subset: Fable 5 45% $\gg$ Opus 25% $>$ Sonnet 20% $\gg$ Haiku 0% top-1; capability-sensitive and far from saturated even for the newest model.



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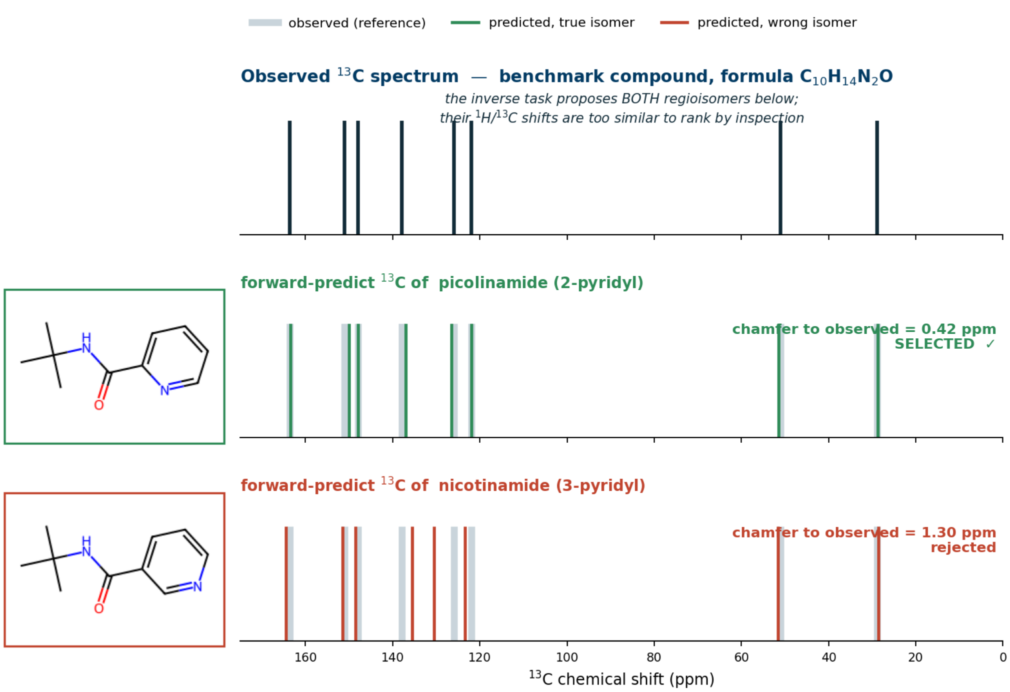

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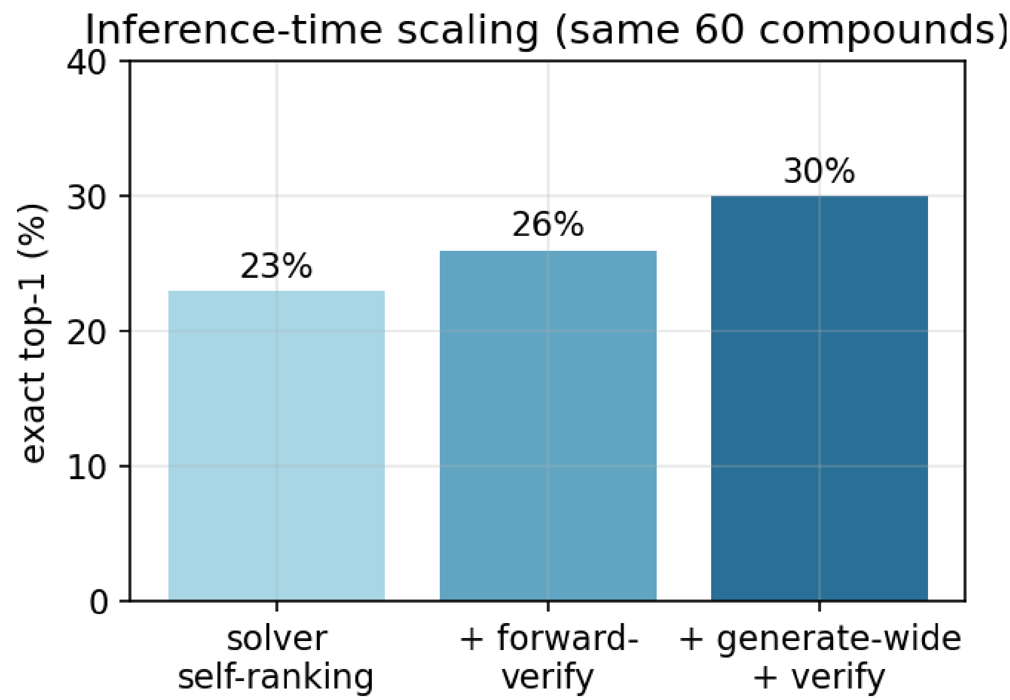

Fig. 8: Forward-verification inference ladder on the same 60 compounds: solver self-ranking → + forward-verify → + generate-wide (23%/26%/30% top-1).